

Journal Recommendation System Using Data Mining Techniques

Data Mining Final Project

Abstract- This project presents a journal recommendation system for computer science research articles. The system takes an article abstract as input and predicts suitable journals for submission. The extracted dataset contained 23,061 article records and 455 publication labels. After removing labels with fewer than 10 samples, the final modeling dataset contained 22,954 records across 402 journal classes. Abstracts were cleaned, converted to TF-IDF vectors, and classified using Logistic Regression. The model achieved 27.55% Top-1 accuracy and 54.24% Top-5 accuracy. K-Means clustering was also applied to generate topic groups such as data mining, software engineering, image processing, security, and networking.

Keywords- Journal recommendation, TF-IDF, Logistic Regression, Top-k accuracy, K-Means clustering, text mining.

I. INTRODUCTION

Selecting an appropriate journal is difficult because many computer science venues cover overlapping topics. Authors often rely on manual search or prior experience, which may be inefficient and subjective. A data mining system can reduce this burden by analyzing an article abstract and recommending journals that are semantically related to the paper.

The objective of this project is to develop a journal finder system tailored to computer science subject areas. The system receives an abstract as input and returns the top five most relevant journals. A second objective is to generate clusters of topics from the article abstracts.

II. DATASET DESCRIPTION

The data was extracted from a relational database. The main tables used were AcademicRecordAbstract, AcademicRecord, and Publication. A SQL join produced pairs of AbstractText and JournalName. The raw dataset contained 23,061 rows and 455 unique journal or publication labels. Journals with fewer than 10 articles were removed, leaving 22,954 records and 402 classes.

Metric	Value
Raw records	23,061
Raw labels	455
Labels removed	53
Final records	22,954
Final classes	402

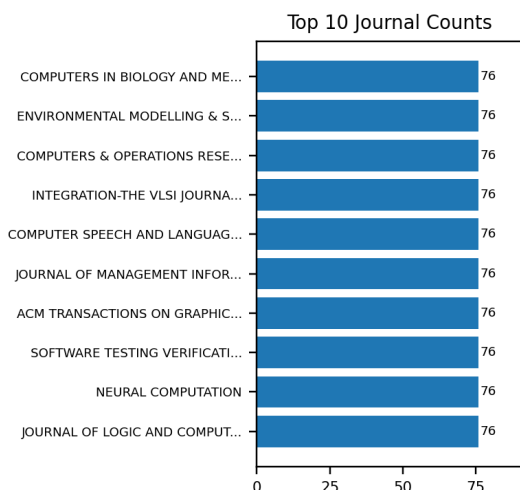


Fig. 1. Distribution of the top 10 journals by number of articles.

III. METHODOLOGY

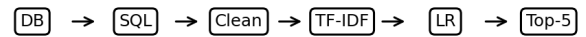


Fig. 2. System architecture.

A. Data Preprocessing

HTML tags were removed from abstracts using regular expressions, and all text was converted to lowercase. Journals with fewer than 10 samples were excluded to avoid unreliable classes.

B. Feature Extraction

TF-IDF vectorization was used to transform text into numerical features. This representation highlights terms that are important to a document while reducing the effect of words that appear across many documents.

C. Classification and Recommendation

Logistic Regression was trained as a multi-class classifier using an 80/20 train-test split. Prediction probabilities were sorted to return the top five journals rather than only one label.

D. Topic Clustering

K-Means clustering with 10 clusters was applied to the TF-IDF abstract vectors. The most important terms near each centroid were used to interpret the topic of each cluster.

IV. RESULTS

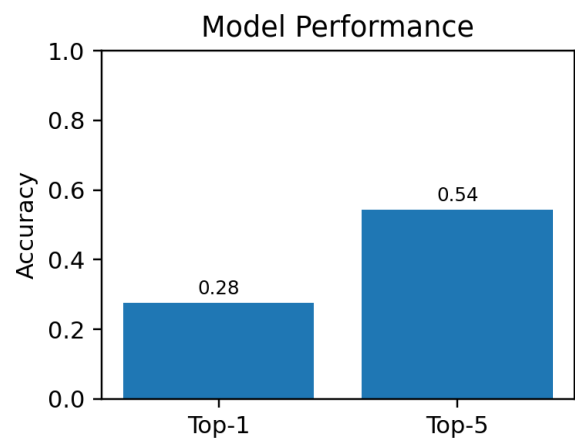


Fig. 3. Top-1 and Top-5 accuracy comparison.

Metric	Result
Top-1 Accuracy	27.55%
Top-5 Accuracy	54.24%
Classes	402
Features	TF-IDF
Classifier	Logistic Regression

The Top-1 accuracy was 27.55%, while the Top-5 accuracy was 54.24%. The Top-5 result is the more important metric because the project requires a ranked recommendation list of five journals.

C	Topic
0	General computational methods
1	Data mining and big data
2	Software engineering
3	Image processing
4	Fuzzy decision systems
5	Information systems
6	Web and cloud services
7	Optimization algorithms
8	Security and cryptography
9	Networks and communications

Table I. Interpreted K-Means topic clusters.

V. DISCUSSION

The Top-1 accuracy is moderate because the classification task contains hundreds of journal classes and many domains overlap. For example, machine learning, optimization, and data mining may appear across several journals. This makes a single exact journal prediction difficult.

The Top-5 accuracy shows stronger practical usefulness. A ranked shortlist is more realistic for journal selection because more than one venue can be suitable for the same abstract. The clustering results also show that TF-IDF captures meaningful topical structure, including security, networks, software engineering, and image processing.

VI. LIMITATIONS AND FUTURE WORK

The current system uses only abstract text. Future work can incorporate keywords, Web of Science subjects, citation statistics, publication year, and journal impact measures. The model can also be compared with Multinomial Naïve Bayes, Support Vector Machines, or transformer-based language models such as BERT.

VII. CONCLUSION

This project developed a journal recommendation system using TF-IDF and Logistic Regression. The system achieved 27.55% Top-1 accuracy and 54.24% Top-5 accuracy across 402 journal classes. K-Means clustering generated meaningful topic groups and satisfied the topic clustering requirement.

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